

May - 21 lecture no: 96

```
```sh
sh kops.sh
```
```

```
```sh
export KOPS_STATE_STORE=s3://reyaz-kops-testbkt143.k8s.local
```
```

```
```sh
kops validate cluster --wait 10m
```
```

```
```sh
kubectl get nodes
```
```

```
```sh
kubectl get namespaces
```
```

```
```sh
kubectl get pods
```
```

```
```sh
cat <<EOF > daemon.yml
apiVersion: apps/v1
kind: DaemonSet
metadata:
 name: mb-daemon
 labels:
 app: bank
spec:
 selector:
 matchLabels:
 app: bank
 template:
 metadata:
 labels:
 app: bank
 spec:
 containers:
 - name: cont1
 image: trainerreyaz/mb-image:latest
EOF
```
```

```
```sh
kubectl get pods
```
```

```
```sh
vi daemon.yml
```
```

```
```sh
kubectl get pods
```
```

```
```sh
kubectl get nodes
```
```

```
```sh
kubectl create -f daemon.yml
```
```

```
```sh
kubectl get pods -o wide
```
```

```
```sh
kubectl delete -f daemon.yml
```
```

```
cat <<EOF > service.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: lb-deployment
  labels:
    app: bank
spec:
  replicas: 3
  selector:
    matchLabels:
      app: bank
  template:
    metadata:
      labels:
        app: bank
    spec:
      containers:
        - name: cont1
          image: trainerreyaz/mb-image:latest
```

```
---
apiVersion: v1
kind: Service
metadata:
  name: lb-service
spec:
  type: ClusterIP
  selector:
    app: bank
  ports:
    - port: 80
      targetPort: 80
EOF
```

```
```sh
kubectl create -f service.yml
```
```

```
```sh
kubectl get svc
```
```

```
```sh
kubectl get pods -o wide
```
```

```
```sh
kubectl get svc
```
```

```
```sh
kubectl describe svc lb-service
```
```

```
```sh
ls daemon.yml kops.sh service.yml
```
```

```
```sh
mv service.yml nodeport.yml
```
```

```
cat <<EOF > nodeport.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: lb-deployment
  labels:
    app: bank
spec:
  replicas: 3
  selector:
    matchLabels:
      app: bank
  template:
    metadata:
      labels:
        app: bank
    spec:
      containers:
        - name: cont1
          image: trainerreyaz/mb-image:latest
---
```

```
apiVersion: v1
kind: Service
metadata:
  name: lb-service
spec:
  type: NodePort
  selector:
    app: bank
  ports:
    - port: 80
      targetPort: 80
      nodePort: 31143
```

EOF

```
```sh
kubectl get svc
```
```

```
```sh
kubectl describe svc lb-service
```
```

```
mv nodeport.yml lb.yml
```

```
kubectl delete -f nodeport.yml
```

Ls

```
cat <<EOF > lb.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: lb-deployment
  labels:
    app: bank
spec:
  replicas: 3
  selector:
    matchLabels:
      app: bank
  template:
    metadata:
      labels:
        app: bank
    spec:
      containers:
      - name: cont1
        image: trainerreyaz/mb-image:latest
---
apiVersion: v1
kind: Service
metadata:
  name: lb-service
spec:
  type: LoadBalancer
  selector:
    app: bank
  ports:
  - port: 80
    targetPort: 80
    nodePort: 31433
```

EOF

```
```sh
kubectl get pods
```
```

```
```sh
kubectl describe svc lb-service
```
```

```
```sh
kops delete cluster --name reyaz.k8s.local --yes
```
```

Lecture 97 : Autoscaling using metrics server

```
```sh
kubectl get nodes
```
```

```
```sh
kubectl apply -f
https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/high-availability-1.21+.yaml
```
```

```
```sh
kubectl top pods
```
```

```
```sh
kubectl autoscale deployment mb-deployment --cpu-percent=20 --min=2 --max=10
```
```

```
```sh
kubectl get pods --watch
```
```

```
```sh
kubectl get hpa
```
```

```
```sh
kubectl get pods
```
```

```
```sh
kubectl exec -it mb-deployment-93893434 -- /bin/bash
```
—
```

```
```sh
apt update
```
```

```
```sh
kubectl exec -it mb-deployment-8585b755c5-b64v5 -- /bin/bash
```
```

```
```sh
stress --cpu 8 --io 4 --vm 2 --vm-bytes 128M --timeout 60s
```
```

```
```sh
kubectl delete -f auto.yml
```
```

```
```sh
cat <<EOF > hpa.yml
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
 name: ib-hpa
spec:
 scaleTargetRef:
 apiVersion: apps/v1
 kind: Deployment
 name: ib-deployment
 minReplicas: 2
 maxReplicas: 10
 metrics:
 - type: Resource
 resource:
 name: cpu
 target:
 type: Utilization
 averageUtilization: 20
EOF
```
```

```
```sh
kubectl exec -it ib-deployment-566f5f5fd6-7kzbp -- /bin/bash
```
```

```
```sh
apt update
```
```

```
```sh
apt install -y stress
```
```

```
```sh
vi hpa.yml
```
```

```
```sh
vi auto.yml
```
```

```
```sh
kubectl create -f auto.yml
```
```

```
```sh
kubectl create -f hpa.yml
```
```

```
```sh
stress --cpu 8 --io 4 --vm 2 --vm-bytes 128M --timeout 60s
```
```

```
```sh
kubectl delete -f auto.yml
```
```

```
```sh
kubectl delete -f hpa.yml
```
```

```
```sh
kubectl get pods
```
```

```
kubectl create ns dev
```

```
kubectl get ns
```

```
Kubectl config view
```

```
kubectl config set-context --current --namespace=dev
```

```
cat <<EOF > dev-quota.yml
apiVersion: v1
kind: ResourceQuota
metadata:
  name: dev-quota
  namespace: dev
spec:
  hard:
    pods: "5"
    limits.cpu: "1"
    limits.memory: 1Gi
EOF
```

```
vi dev-quota.yml
```sh
vi dev-quota.yml
```

```
'''
```

```
```sh
```

```
kubectl create -f dev-quota.yml
```

```
'''
```

```
```sh
```

```
kubectl get quota
```

```
'''
```

```
cat <<EOF > deploy.yml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
 name: ib-deployment
```

```
 labels:
```

```
 app: bank
```

```
spec:
```

```
 replicas: 3
```

```
 selector:
```

```
 matchLabels:
```

```
 app: bank
```

```
 template:
```

```
 metadata:
```

```
 labels:
```

```
 app: bank
```

```
 spec:
```

```
 containers:
```

```
 - name: cont1
```

```
 image: reyadocker/internetbankingrepo:latest
```

```
 resources:
```

```
 limits:
```

```
 cpu: "1"
```

```
 memory: 512Mi
```

```
EOF
```

```
kubectl create -f deploy.yml
```

```
Kubectl get pods
```

```
```sh
```

```
kubectl delete -f deploy.yml
```

```
'''
```

```
cat <<EOF > deploy.yml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: ib-deployment
```

```
  labels:
```

```
  app: bank
spec:
  replicas: 3
  selector:
    matchLabels:
      app: bank
  template:
    metadata:
      labels:
        app: bank
    spec:
      containers:
        - name: cont1
          image: reyadocker/internetbankingrepo:latest
          resources:
            limits:
              cpu: "0.3"
              memory: 300Mi
```

EOF

Kubectl delete -f deploy.yml

Kubectl get pods